

Immigration and Credit in America

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- Immigrants start life in the U.S. *without* a U.S. credit report
 - any foreign credit history generally unobserved to U.S. lenders

How do immigrants assimilate into the formal U.S. consumer credit system?

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Data

- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
- Data **matched** with information on whether a consumer is an **immigrant**, & **year of immigration**, using sequential Social Security Number (SSNs) assignment

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What We Do

- Describe **selection** of immigrants compared to non-immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration, conditional on birth year & geographic fixed effects

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What We Do

- Describe **selection** of immigrants & **assimilation dynamics** over life cycle
 - Compare immigrants vs. non-immigrants, immigrants arriving at different ages

Key Findings. Immigrants vs. non-immigrants, & immigrants arriving at older ages:

- **Positive selection:** More creditworthy (once credit scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans & mortgages, convergence in credit card access
- Credit card limits are **lower** but become **higher over time**

Our paper contributions to literatures across labor & household finance

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Sequeira, Nunn, & Qian, 2020; Kerr & Kerr, 2020; Azoulay, Jones, Kim, & Miranda, 2022; Bernstein, Diamond, McQuade, Jiranaphawiboon, & Pousada, 2025)
- We show the selection of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1987; Abramitzsky & Boustan, 2017)
- We show assimilation into the U.S. economy over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzsky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Geber, & Isen, 2022)
- We contribute to the household finance literature on economics of credit information, & credit use across the life cycle that largely has not studied role of immigrants (e.g., Pagano & Jappelli, 1996; Brevoort et al., 2015, 2017, 2018; Nathe, 2021; Blattner, Hartwig, & Nelson, 2022; Zillessen, 2022; Bach et al., 2023; Ricks & Sandler, 2025)

1. Data & Immigrant Classification
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

Data & Immigrant Classification

US Consumer Credit Reporting Data (BTCCP)

- University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)
 - Anonymized data sample of 1 in 10 consumers with US credit reports
 - Monthly data from 2000 to 2024. Includes:
 - Anonymized consumer identifiers
 - Individual tradeline data (e.g., details on each account, including opening dates)
 - Consumer-level data (e.g., birth date, credit score, ZIP code)
- (Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang, 2025)

Classifying Immigrants

Before 2012, Social Security Administration assigned Social Security Numbers (SSNs) sequentially within states

Consumers assigned SSNs as adults very likely to be (legal) immigrants

- validated in prior work (Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein, Diamond, McQuade, Jiranaphawiboon, & Pousada, 2025)
- **“Immigrants”**: assigned SSN at age 21+. **“Non-immigrants”**: assigned SSN at age < 21

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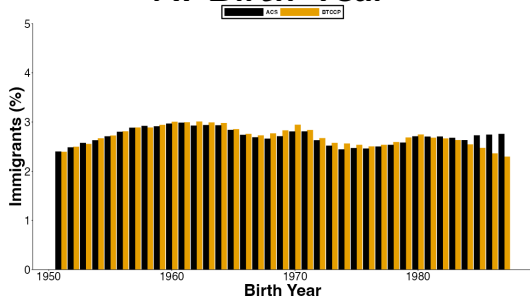
Classifying Immigrants in BTCCP

1. Created lookup table: map first 5 digits of SSN to SSN assignment year (*SSN Year*)
2. Sent to TransUnion: (i) lookup table (ii) list of anonymous consumer identifiers in BTCCP
3. TransUnion merged to their non-anonymous data that includes SSNs. TransUnion returned consumer SSN Year (without SSNs) linked to anonymous consumer identifiers
4. We calculate age of SSN assignment: $SSN\ Age = SSN\ Year - Birth\ Year$

Our Immigrant Classification Aligns with American Community Survey (ACS)

Distribution of Immigrants by...

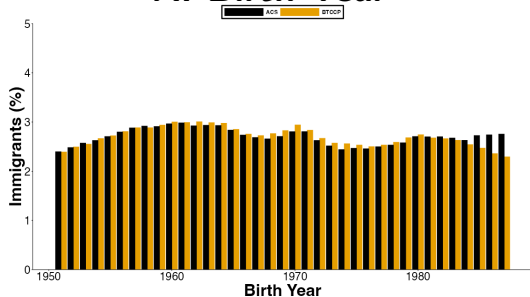
A. Birth Year



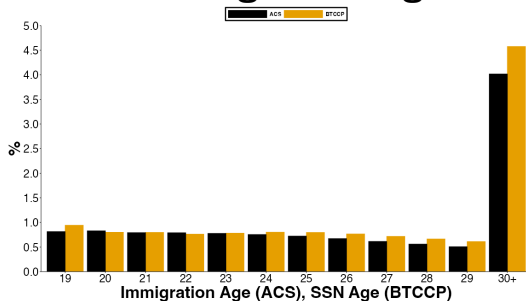
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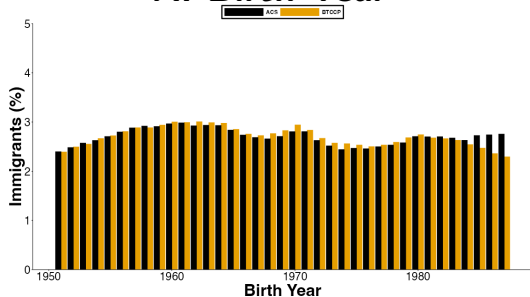
B. Immigration Age



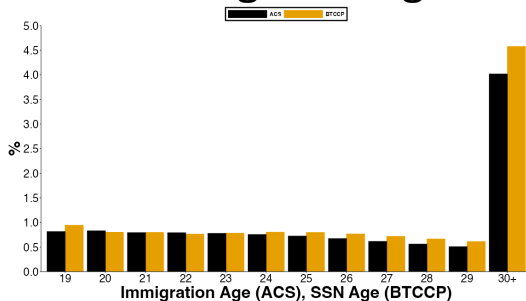
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Distribution of Immigrants by...

A. Birth Year



B. Immigration Age



Distribution of immigrants also aligns by geography

Immigrants more geographically mobile than non-immigrants

“Entrant Sample”: Restrict to Birth Years 1975 - 1987 to Observe U.S. Entry in Ages 21 to 29 & Credit Outcomes for Ages 18 to 40

TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
 Clean Sample	 18,572,654
... SSN Age <21	16,478,940
... SSN Age 21+	2,093,714 (11.27%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
 Entrant Sample (SSN Age < 30)	 6,122,932
... SSN Age < 21	5,778,671
... SSN Age 21-29	344,261 (5.96%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

Institutional Details on Lending in America

- Immigrant credit access may be adversely affected by:
 1. **Not having a credit report on arrival**
 - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
 2. **After entering the credit system, having a shorter history than non-immigrants**
 - Length of credit report has positive impact to credit scores ($\approx 15\%$ FICO)
 3. **Other cultural barriers**
 - Differences in their familiarity & willingness to use debt, how to build credit score

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 3. **Other cultural barriers**
 - Differences in their familiarity & willingness to use debt, how to build credit score
- Lenders use credit scores with other information in lending decisions
 - Lenders can use immigration status
 - Lenders cannot use protected characteristics (e.g., gender, national origin, race) under Equal Credit Opportunity Act (ECOA)

Credit Market Entry

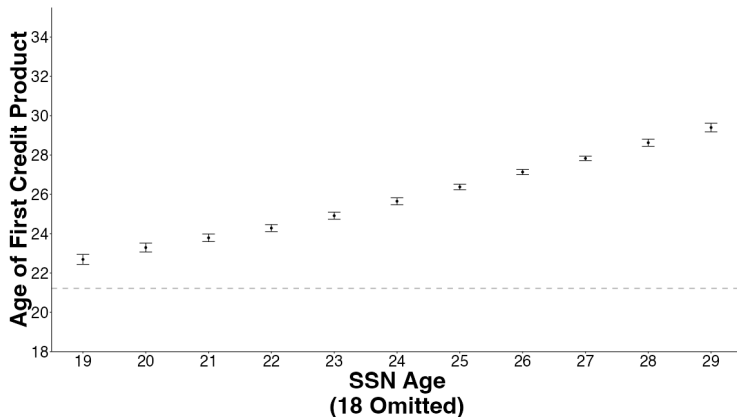
Cross-Sectional OLS Regression:

$$Y_i = \phi_{SSN} Age(i) + \mu_{b(i)} + \epsilon_i \quad (1)$$

Cluster standard errors by birth year

SSN Assigned at Older Age → Later Entry into American Credit System

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \quad (1)$$



Main Cross-Sectional Regression Specification

Cross-Sectional OLS Regression:

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \quad (1)$$

Given **linearity** in *SSN Age*, we mainly use simpler OLS cross-sectional regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \quad (2)$$

β_1 : average difference for immigrants vs. non-immigrants

β_2 : marginal difference for SSN Age one year later

Cluster standard errors by birth year

Results robust to adding geographic fixed effects to control

(First observed ZIP5, Last observed ZIP5, Longest-held ZIP5, number of ZIP5s)

Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later than Non-Immigrants & Younger SSN Age Immigrants

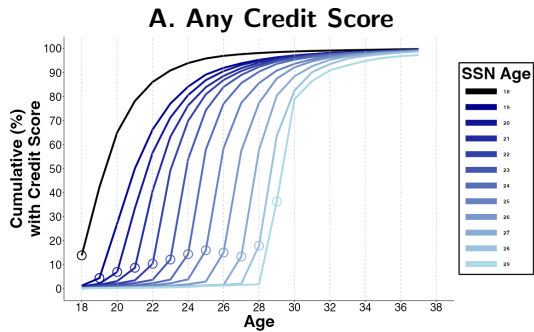
$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \quad (2)$$

Age at First...	Credit Report (1)	Credit Product (2)
21+	1.972*** (0.101)	1.665*** (0.112)
SSN Age	0.740*** (0.015)	0.704*** (0.017)
Birth Year F.E.	X	X
R^2	0.267	0.087
Mean, SSN Age <21	19.86	21.25

$N = 6,122,932$

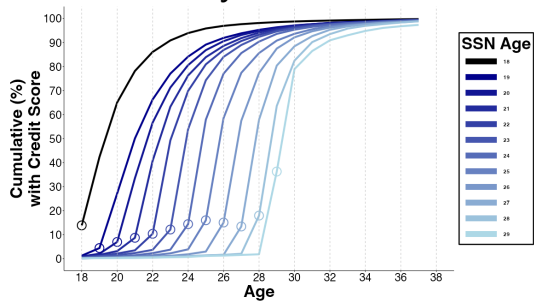
Creditworthiness

Credit Score

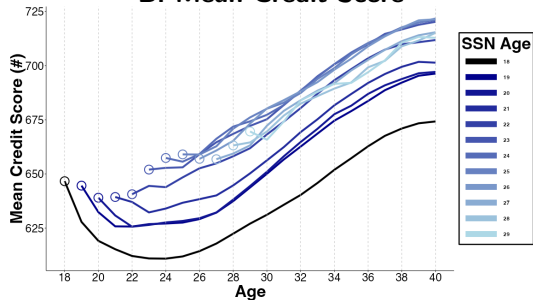


Credit Score

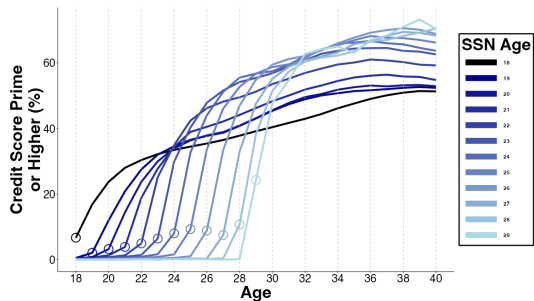
A. Any Credit Score



B. Mean Credit Score

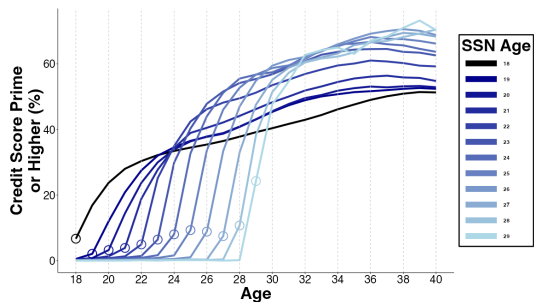


C. Prime or Higher Credit Score

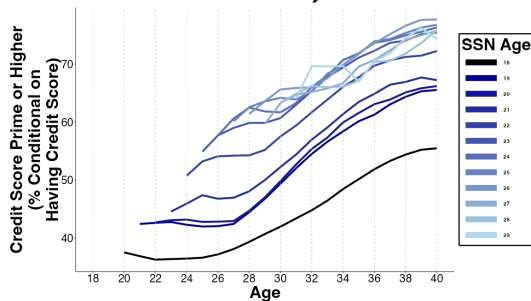


Positive Selection of Immigrants in Credit Score Distribution

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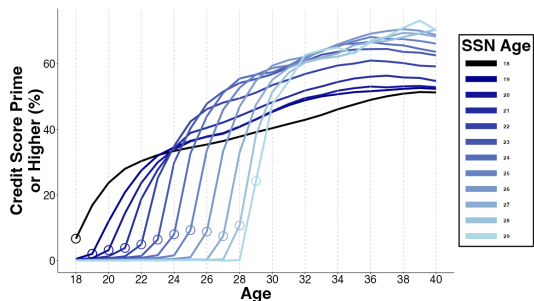


D. Prime or Higher Credit Score (Conditional on Any Score)

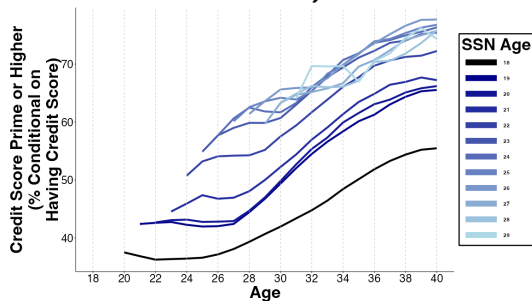


Positive Selection of Immigrants in Credit Score Distribution

C. Prime or Higher Credit Score



D. Prime or Higher Credit Score (Conditional on Any Score)



Immigrants (vs. Non-Immigrants) have more compressed credit scores:

$\downarrow \Pr(\text{subprime}), \uparrow \Pr(\text{superprime})$

Immigrant Credit Scores Higher In Spite Of Shorter Credit Histories

	Credit Score at		Any Prime or Higher (660+) Credit Score at	
	Age 30	Age 40	Age 30	Age 40
	(1)	(2)	(3)	(4)
21+	23.7*** (2.1)	32.3*** (0.7)		
SSN Age	3.2*** (0.4)	2.4*** (0.2)		
Birth Year F.E.	X	X		
R^2	0.009	0.022		
N	5,755,134	4,449,929		
Mean, SSN Age <21	626.1	659.5		

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \quad (2)$$

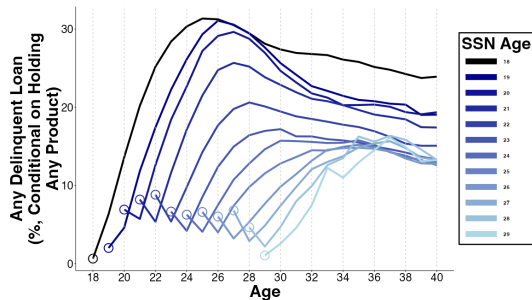
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	Credit Score at		Any Prime or Higher (660+) Credit Score at	
	Age 30	Age 40	Age 30	Age 40
	(1)	(2)	(3)	(4)
21+	23.7***	32.3***	9.50***	4.08***
	(2.1)	(0.7)	(0.83)	(0.29)
SSN Age	3.2***	2.4***	0.29	1.92***
	(0.4)	(0.2)	(0.23)	(0.10)
Birth Year F.E.	X	X	X	X
R^2	0.009	0.022	0.006	0.0100
N	5,755,134	4,449,929	6,122,932	4,800,195
Mean, SSN Age <21	626.1	659.5	37.71	46.84

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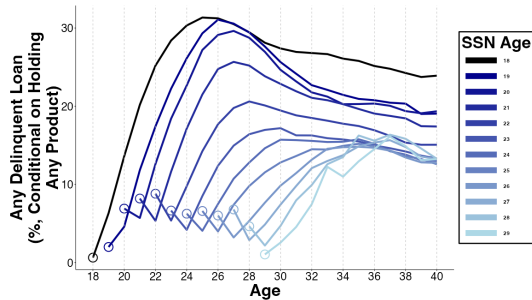
Immigrants (vs. Non-Immigrants) have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency

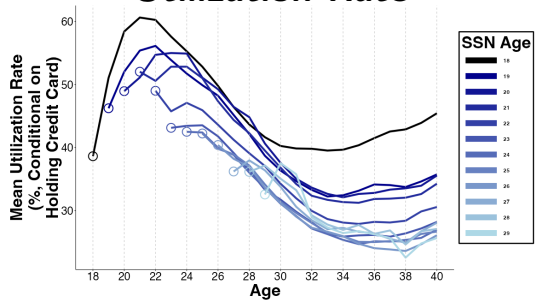


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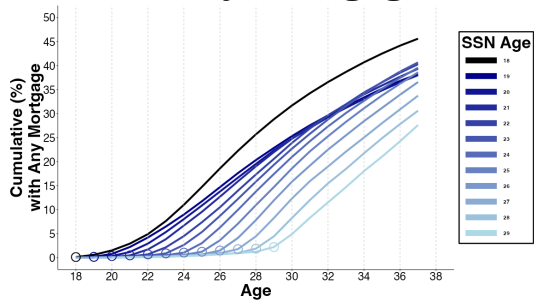
B. Credit Card Utilization Rate



Credit Access

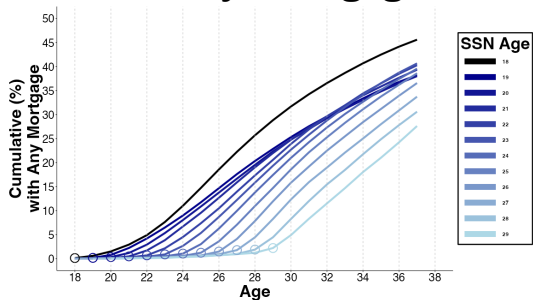
Immigrants Persistently Less Likely to Hold Mortgages & Auto Loans

A. Any Mortgage

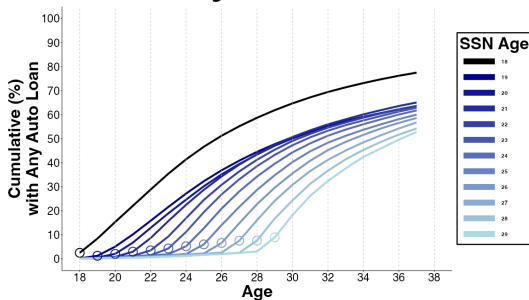


Immigrants Persistently Less Likely to Hold Mortgages & Auto Loans

A. Any Mortgage



B. Any Auto Loan

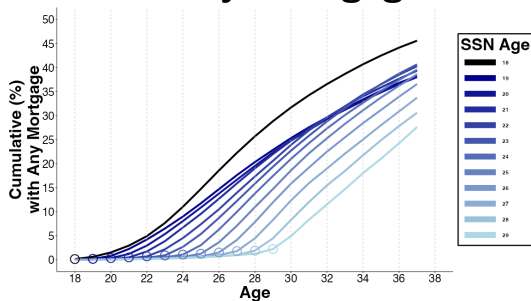


By Age 37, Immigrants are:

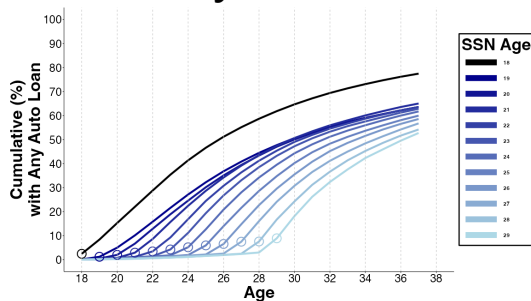
- 2.3 (s.e. 0.6) p.p.(5%) *less* likely to have mortgage...but *not* robust
- Each SSN Age year is 1.6 (0.1) p.p. *less* likely to have mortgage

Immigrants Persistently Less Likely to Hold Mortgages & Auto Loans

A. Any Mortgage



B. Any Auto Loan



By Age 37, Immigrants are:

2.3 (s.e. 0.6) p.p.(5%) *less* likely to have mortgage...but *not* robust

- Each SSN Age year is 1.6 (0.1) p.p. *less* likely to have mortgage

10.2 (s.e. 0.4) p.p. (13%) *less* likely to have auto loan

- Each year older SSN Age, 1.4 (0.1) p.p. *less* likely to have auto loan

Immigrants, & Older SSN Age Immigrants, Only First Access Mortgage, Auto Loan, & Credit Card at Older Ages

$$Y_i = \beta_1 \cdot \mathbb{I}\{\text{SSN Age } 21+\}_i + \beta_2 \cdot \text{SSN Age}_i + \mu_{b(i)} + \epsilon_i \quad (2)$$

Age at First...	Mortgage (1)	Auto Loan (2)	Credit Card (3)
21+	0.781*** (0.097)	2.930*** (0.217)	1.160*** (0.169)
SSN Age	0.403*** (0.018)	0.612*** (0.023)	0.663*** (0.022)
Birth Year F.E.	X	X	X
R^2	0.024	0.030	0.033
Mean, SSN Age <21	36.36	29.15	22.68

$N = 6,122,932$

“Paired Cohorts”: Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{focal\ cohort\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- ‘Focal Cohort’: $SSN\ Age_c$, 22-30
- ‘Control Cohort’: $SSN\ Age_c - 1$, same birth year. 68 pairs (p)
- 2,716 cohort-by-year observations ($k(p), t$), s.e. clustered by birth year

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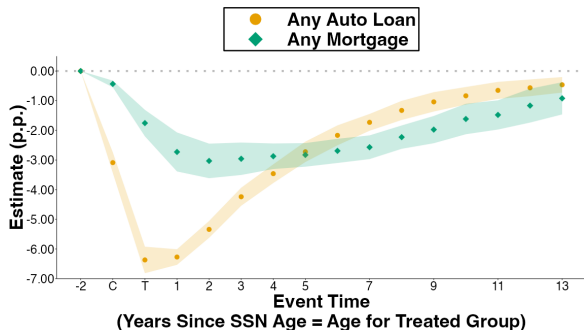
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- **Results:**

Any Auto Loan after 10 years:

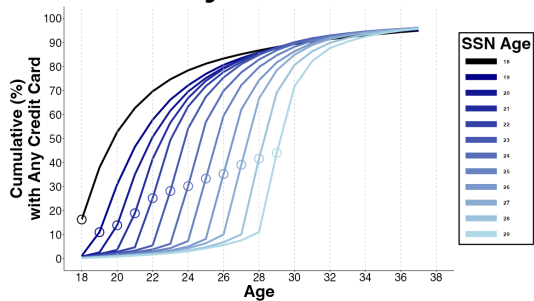
−0.46 p.p., −0.42 cumulative years

Any Mortgage after 10 years:

−0.92 p.p., −0.29 cumulative years

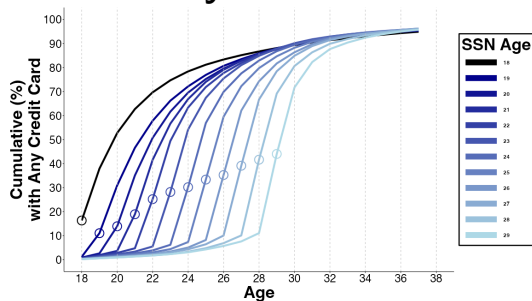


A. Any Credit Card

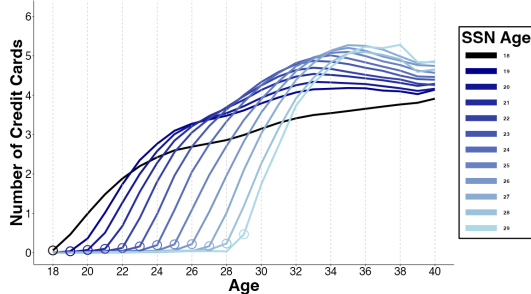


Immigrants' Access to Credit Cards Converges

A. Any Credit Card



B. Number of Credit Cards



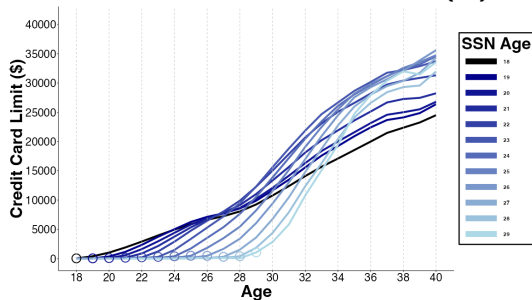
By Age 37, Immigrants are:

0.97 (s.e. 0.16) p.p. (1%) *more* likely to have accessed credit card

- No robust result for older SSN Ages affecting credit card use

Immigrants' Credit Card Limits Are Initially Lower, But Become Higher than Non-Immigrants

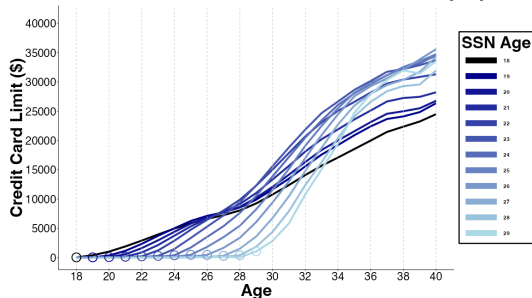
A. Credit Card Limits (\$)



Similar pattern for credit card limit per card

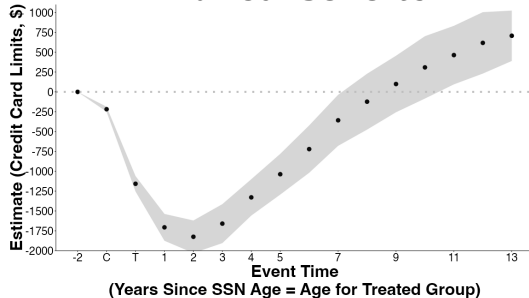
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A. Credit Card Limits (\$)



Similar pattern for credit card limit per card

B. Paired Cohorts



SSN Age +2 Years:

-\$1,825 (s.e. \$105)

SSN Age +13 Years:

+\$707 (s.e. \$162)

Mechanisms

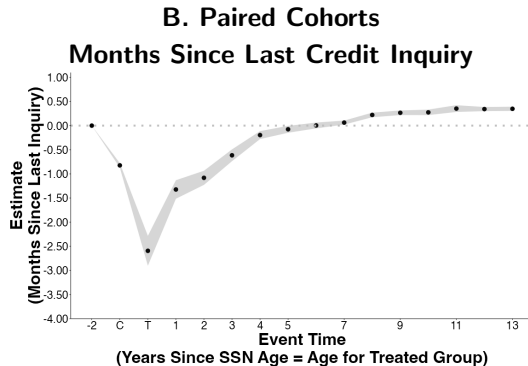
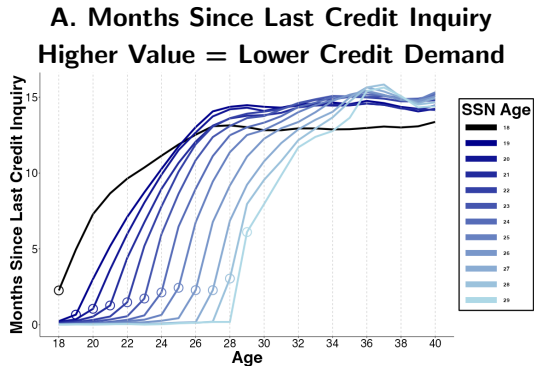
Immigrants vs. non-immigrants, & immigrants arriving at older (vs. younger) ages are:

- **More creditworthy** (once credit scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans & mortgages, convergence in credit card access
- Credit card limits are **lower** but become **higher over time**

1. **Creditworthiness** ✖
 - Positive selection of immigrants, & immigrants arriving later in life
2. **Credit demand**
3. **Emigration**
4. **Tastes**
5. **Information friction**

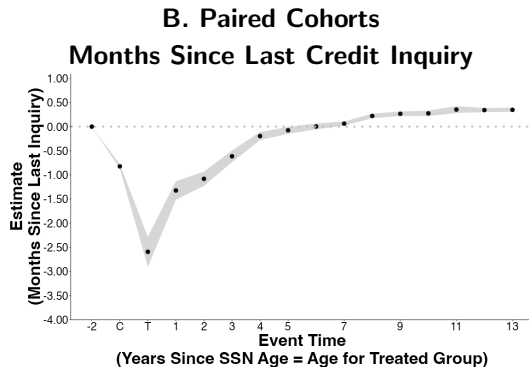
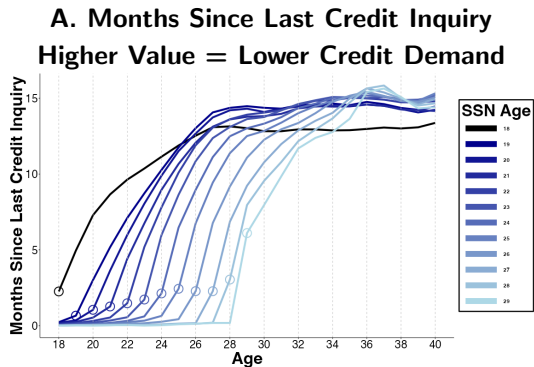
Potential Mechanism #2: Does Lower Credit Demand Explain Lower Access?

Immigrants have higher credit demand in 20s, lower demand in 30s



Potential Mechanism #2: Does Lower Credit Demand Explain Lower Access?

Immigrants have higher credit demand in 20s, lower demand in 30s



Caveat: Lower credit demand in 30s may be function of being earlier rejections → deterrence

Potential Mechanism #3: Emigration

Proxy emigration by **attrition** from data

Immigrants exit data more than non-immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

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Restrict to consumers observed in 2024:

- **Mortgage** immigrant vs. non-immigrant gap closes but SSN Age gap remains
- **Auto loan** immigrant vs. non-immigrant gap & SSN Age gap remains
- **Credit card** dynamics by SSN Age remains

Potential Mechanism #4: Tastes

Differences between immigrants and non-immigrants may be explained by tastes:

- Immigrants have different cultural backgrounds, may be less familiar/willing to use debt
- Autos & Property: Choose not to purchase / purchase without debt

Potential Mechanism #4: Tastes

Differences between immigrants and non-immigrants may be explained by tastes:

- Immigrants have different cultural backgrounds, may be less familiar/willing to use debt
- Autos & Property: Choose not to purchase / purchase without debt

Tastes unlikely to explain SSN Age gaps or paired cohorts results

- One potential taste-based explanation, immigrant's life cycle less geared around birth year, & more around time since immigrating

Potential Mechanism #5: Information Friction

Information friction of non-portability of credit histories across national boundaries may prevent profitable lending. **Solutions?**

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American Express Global Card Relationship

Global backing for Card Members,
wherever you move

Global Card Relationship, our Card Member-only service, makes it seamless for Card Members to apply for a new American Express® Card in select countries.

Nova Credit

International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

Equifax Canada Global Consumer Credit File

Global Consumer Credit File

Helping you grow and retain your clients
who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

Potential Mechanisms

1. Creditworthiness ✖
2. Credit demand ✖
3. Emigration ✔ ✖
4. Tastes ?
5. Information friction ?

Conclusions

How do immigrants assimilate into the formal U.S. consumer credit system?

Study **selection & assimilation dynamics** over life cycle of immigrants vs. non-immigrants, & exploit variation in age of immigration, using credit data matched with immigration information

Key Findings. Immigrants (vs. non-immigrants), & immigrants arriving at older ages:

- Are **positive selected**: more creditworthy (once scored)
- Experience **delayed entry** into credit system
- **Lower long-term access** to auto loans & mortgages, convergence in credit card access
- Credit card limits are **lower** but become **higher over time**
- More work to do on **mechanisms**, but our main results are **not explained** by **creditworthiness, credit demand, or emigration**

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Implication is that immigrants are profitable to lend to. Potential solutions?

- **FinTech innovations** (e.g., AmEx Global Card Relationship, NovaCredit, Global Credit File)
- **Removing barriers** to lending to immigrants or sharing information across-borders
- **Shorten duration** for retaining positive information on credit reports, but this has trade-offs (e.g., Blattner, Nelson, & Hartwig, 2022; Kovbasyuk & Spagnolo, 2024)

Thank you!

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